How a page is retrieved for a store manager who sees 100 of 1,000,000 products

Input: the caller's principal ids, the permission "product.view", the query (Products sorted by Price), the previous cursor, page size 20.
Two small statements resolve the principals (the user and her groups) and the permission. Nothing about her grants leaves the database.

1. Scan step

Read the next 20 rows of the query in price order.
Tag each with fn_IsResourceAccessible in the same statement.

Price	Id	visible?
0.01	p48811	no
0.02	p1092	no
0.03	p77250	no
...	17 more rows	no

Result: 0 of 20 visible. An administrator would stop here with 20 visible.
Cost: 20 rows read, 20 access checks, whoever the caller is.

20 visible rows (dense caller)

2. Merge step (only when an index covers the order)

fn_PageNodes(principals, permission) returns her minimal granted nodes, computed inside the database: here one node, her store S7.

SqlOSFgaPage_Products_Price, one index seek per node:

NodeId	Price	RowKey
S6	...	...
S7	4.10	p3304
S7	4.95	p3417
S7	5.20	p3350
S7	... 17 more, in price order	
S8	...	...

Under each node: the next 20 entries after the cursor that pass the query's filters. Merge the nodes' ranges by price and keep the first 20 keys (CROSS APPLY / JOIN LATERAL).

Result: 20 keys. Cost: one seek per node, about 20 entries each.
A caller with 100 stores: 100 seeks. The table's size never appears.

20 keys (sparse caller)

3. Load and final check

Load the 20 rows through the application's own query, with its includes and change tracking,
and apply fn_IsResourceAccessible once more to each row. A stale copy entry can hide a row; it can never return one the caller may not see.

Next cursor: the price and id of the last row returned. The next page starts from there, in either step.

Output: a page of 20 rows and a cursor. Rows read for this caller: 80 (20 scanned, 40 in the merge step, 20 loaded).
Under the per-row check alone: 184,205 rows.

The scan step runs first because it costs twenty checks and settles the page for every caller who sees most of the table, without touching a copy.